

Lecture 4:

Predicate Calculus

Predicates

- $C x = x \text{ is a cat}$
 - $S x = x \text{ is a scientist}$
 - $g = \text{Galileo}$
 - $a = \text{Apollo}$
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- $S g = \text{Galileo is a scientist}$
 - $C a = \text{Apollo is a cat}$

Predicates (cont)

- $F x y = x \text{ is more famous than } y$
- $P x y z = x \text{ is playing } y \text{ in } z$
- $P b v m = \text{Boris is playing the violin in lake michigan}$
- $F g a = \text{Galileo is more famous than Apollo}$

Universal quantification (Forall)

- $\forall x, P$
- $\forall x, S x$
- $\forall x, S x \rightarrow \neg C x$
- $\forall x, \forall y, S x \wedge C y \rightarrow F x y$
- $\forall y, \forall x, S x \wedge C y \rightarrow F x y$

Existential Quantification (exists)

- $\exists x, P$
- $\exists x, C x$
- $\exists x, S x \wedge \exists y, F y x$
- $\exists x, \exists y, C y \wedge S x \wedge F y x$

Combining Quantifiers

- $\forall x, \exists y, Cx \rightarrow Fyx$
- $\exists y, \forall x, Cx \rightarrow Fyx$

- $\forall x, \exists y, x + y = 5$
- $\exists x, \forall y, x + y = 5$
- $\exists x, \forall y, x * y = x$

Forall Rules

For *arbitrary* a

$$\frac{Pa}{\forall x, Px} \quad \forall I$$

$$\frac{\forall x, Px}{Pa} \quad \forall E$$

For *any* a

$$Pa \vee \neg Pa$$

Exists Rules

For *some* a

$$\frac{Pa}{\exists x, Px} \quad \exists\text{-I}$$

$$\frac{\exists x, Px}{Pa} \quad \exists\text{-E}$$

For *fresh* a

Derivation

- $S\ x = x$ is a famous scientists
- $C\ x = x$ is a Cat
- $N\ x\ y = x$ is named after y
- $g = \text{Galileo Galilei}$
- Assumption: $S\ g$
- Assumption: $\forall\ x, S\ x \rightarrow \exists\ y, C\ y \wedge N\ y\ x$
- Conclusion: $\exists\ x, N\ x\ g$

$S x = x$ is a famous scientists

$C x = x$ is a Cat

$N x y = x$ is named after y

$$\begin{array}{c} \forall x, S x \rightarrow \exists y, C y \wedge N y x \\ \hline S g \rightarrow \exists y, C y \wedge N y g \qquad S g \\ \hline \exists y, C y \wedge N y g \\ \hline C c \wedge N c g \\ \hline N c g \\ \hline \exists x, N x g \end{array}$$